PROJECT REPORT

TITLE: MOTION TRIGGERED VIRTUAL PIANO

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COURSE: FUNDAMENTALS IN AIML

BRANCH : AI AND ML

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1. Introduction-

The Virtual Piano is a real-time computer vision application that allows users to play musical notes by moving their hands in front of a webcam. This project

eliminates the need for physical hardware or touch interfaces, using motion tracking and color segmentation to trigger audio frequencies.

2. Objectives-

- To develop a touchless interface for musical expression.
- To implement real-time skin color detection using the HSV color space.
- To map spatial coordinates of detected objects to specific musical frequencies (261.63 Hz to 523.25 Hz).

3. Methodology & Technical Implementation-

The system follows a linear pipeline of image processing and decision-making:

A. Video Capture and Preprocessing-

- Frame Acquisition: The system captures video at the hardware's default resolution using cv2.VideoCapture.
- Mirroring: The frame is flipped horizontally using cv2.flip(frame, 1) to provide a natural "mirror" interaction for the user.
- Color Conversion: Frames are converted from BGR to HSV to improve the robustness of skin detection under varying lighting.

B. Hand Detection (Skin Masking): The core interaction relies on a skin-tone mask defined by specific HSV boundaries:

- Lower Boundary: [0, 40, 60]
- Upper Boundary: [20, 255, 255]

Any pixels falling within this range are converted to a white mask, while all others become black.

C. Spatial Mapping (The "Hot Zone"):-

- Region of Interest (ROI): The system only monitors the top 60 pixels of the frame.
- Contour Analysis: The largest white object in the ROI is identified as the hand.
- Centroid Calculation: The x-coordinate of the hand is calculated using Image Moments (M_{10}/M_{00}).

D. Audio Synthesis: The screen is divided into 8 equal sections.The calculated x-coordinate determines which "key" is active.The winsound.Beep function generates the tone based on the FREQUENCIES array.

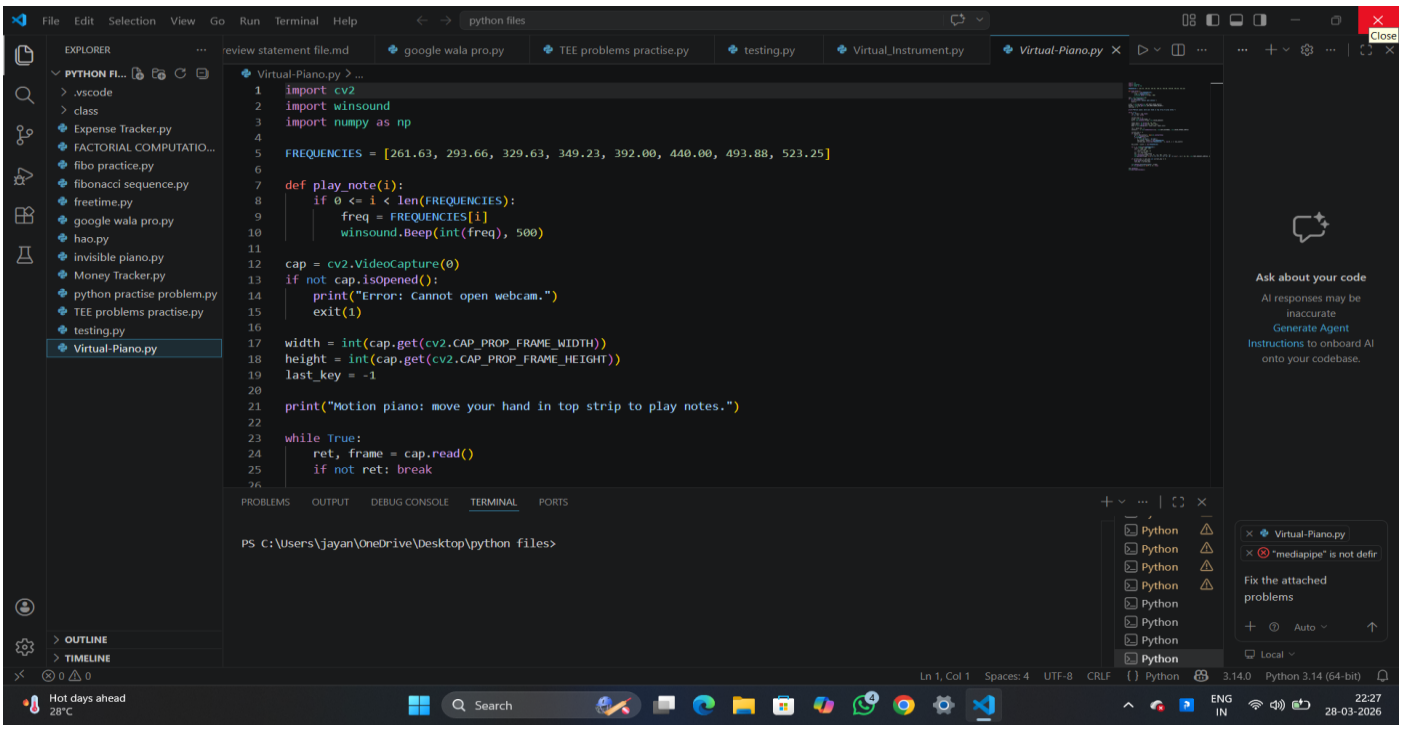
4. Hardware and Software Requirements-

Component	Specification
Operating System	Windows (for winsound support)
Language	Python 3.x
Libraries	OpenCV (cv2), NumPy, winsound
Hardware	Standard USB or Integrated Webcam

5. Results and Observations-

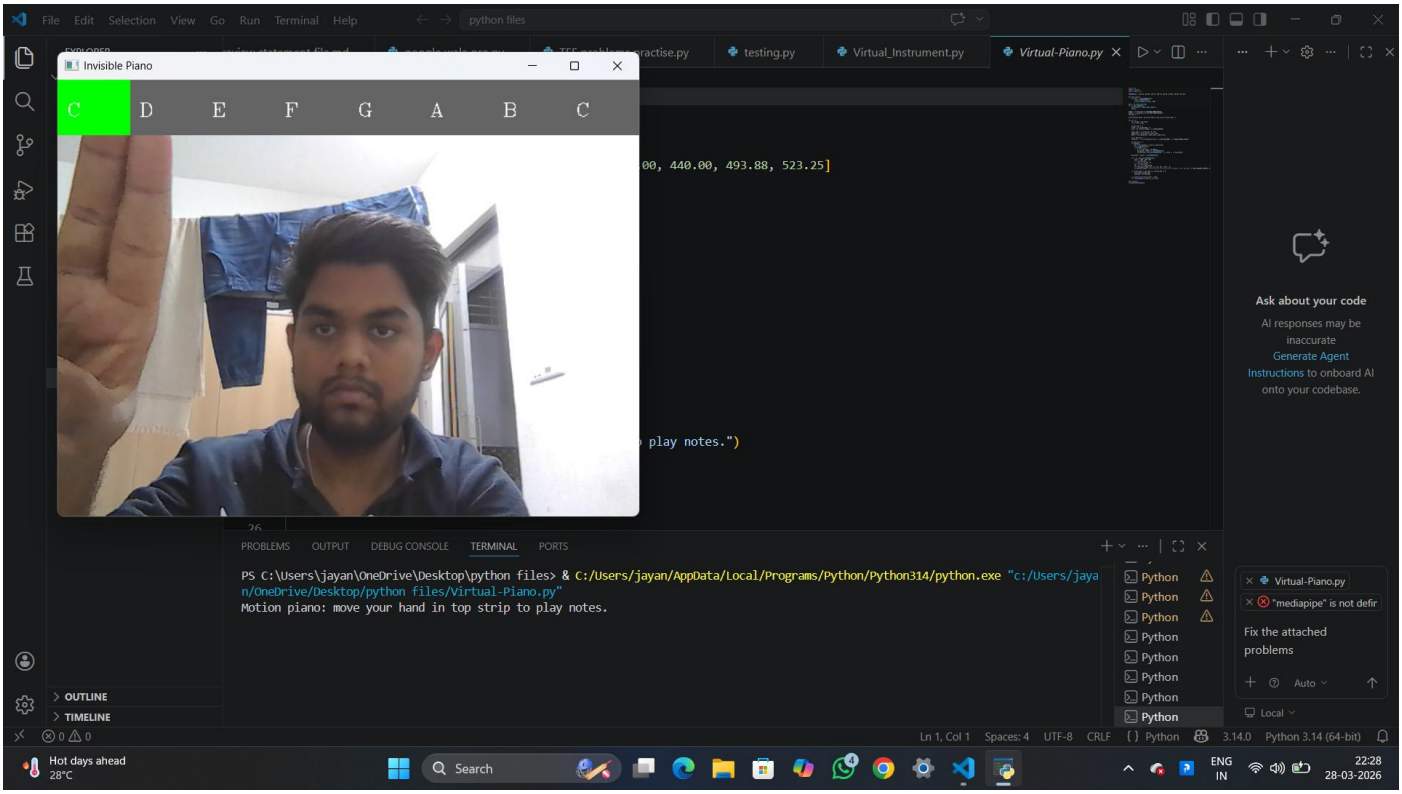
- Responsiveness: The system triggers notes immediately upon the hand entering a new segment.
- Visual Feedback: The interface draws green rectangles over the active "key" to confirm user input.
- Limitations: Performance is highly dependent on lighting; dark environments may result in "noisy" detection or failure to trigger notes.

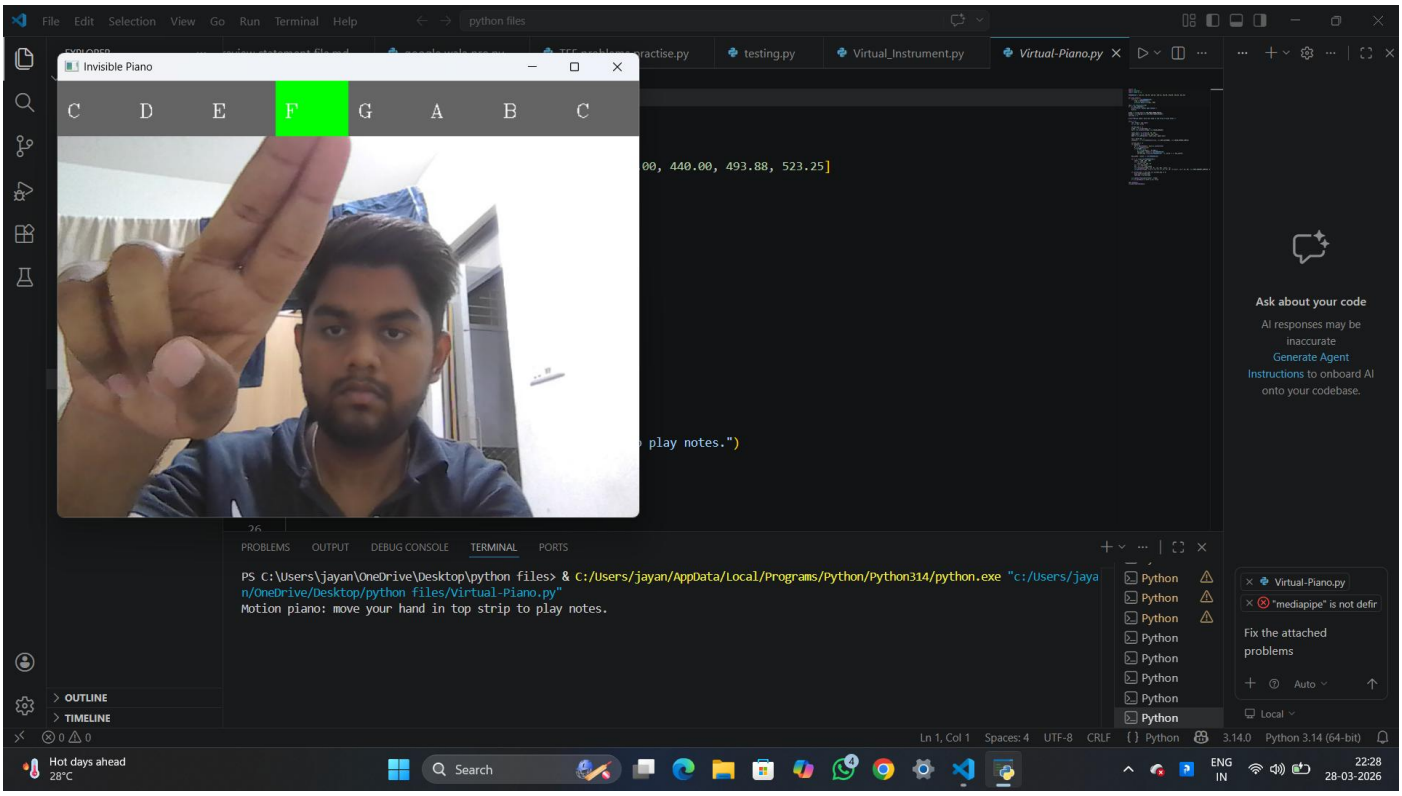
Screenshots:



```python
    current_key = -1
    frame = cv2.flip(frame, 1)
    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

    lower_skin = np.array([0, 40, 60])
    upper_skin = np.array([20, 255, 255])
    mask = cv2.inRange(hsv, lower_skin, upper_skin)

    roi = mask[:60, :]
    contours, _ = cv2.findContours(roi, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    current_key = -1
    if contours:
        cnt = max(contours, key=cv2.contourArea)
        M = cv2.moments(cnt)
        if M["m00"] > 0:
            x = int(M["m10"] / M["m00"])
            key_width = width // len(FREQUENCIES)
            current_key = min(len(FREQUENCIES) - 1, max(0, x // key_width))

    key_width = width // len(FREQUENCIES)

    for i in range(len(FREQUENCIES)):
        color = (100, 100, 100)
        if i == last_key:
```

```python
    key_width = width // len(FREQUENCIES)

    for i in range(len(FREQUENCIES)):
        color = (100, 100, 100)
        if i == last_key:
            color = (0, 255, 0)
        x1 = i * key_width
        x2 = (i + 1) * key_width
        cv2.rectangle(frame, (x1, 0), (x2, 60), color, -1)
        cv2.putText(frame, f"{['C','D','E','F','G','A','B','C'][i]}", (x1 + 10, 40), cv2.FONT_HERSHEY_COMPLEX, 0.7, (255, 255, 255),

    if current_key != last_key and current_key >= 0:
        play_note(current_key)
        last_key = current_key

    cv2.imshow("Invisible Piano", frame)
    if cv2.waitKey(1) & 0xFF == 27: break

cap.release()
cv2.destroyAllWindows()
```


Invisible Piano
C D E F G A B C
PS C:\Users\jayan\OneDrive\Desktop\python files> & C:/Users/jayan/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/jayan/OneDrive/Desktop/python files/Virtual-Piano.py"
Motion piano: move your hand in top strip to play notes.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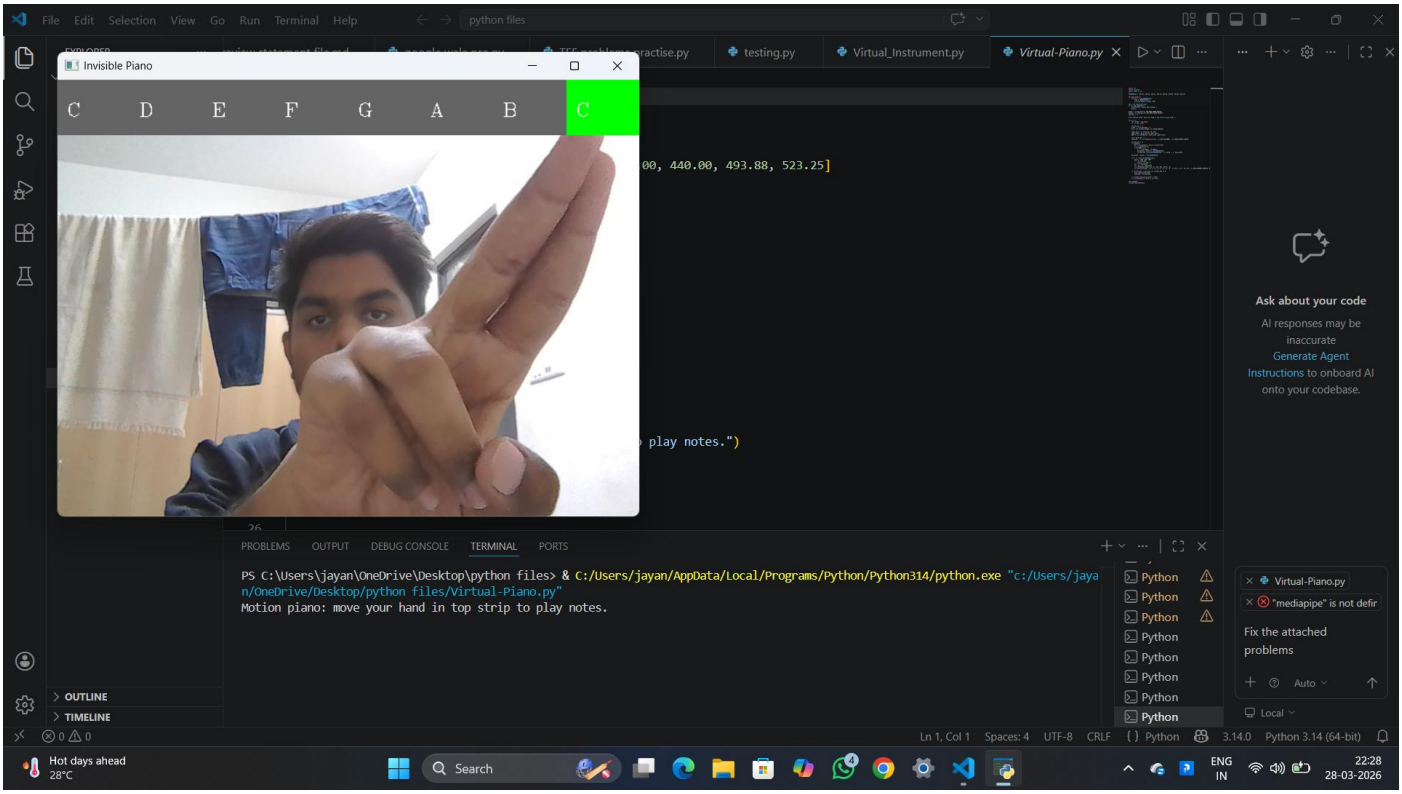
Invisible Piano
C D E F G A B C
00, 440.00, 493.88, 523.25]
play notes.")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
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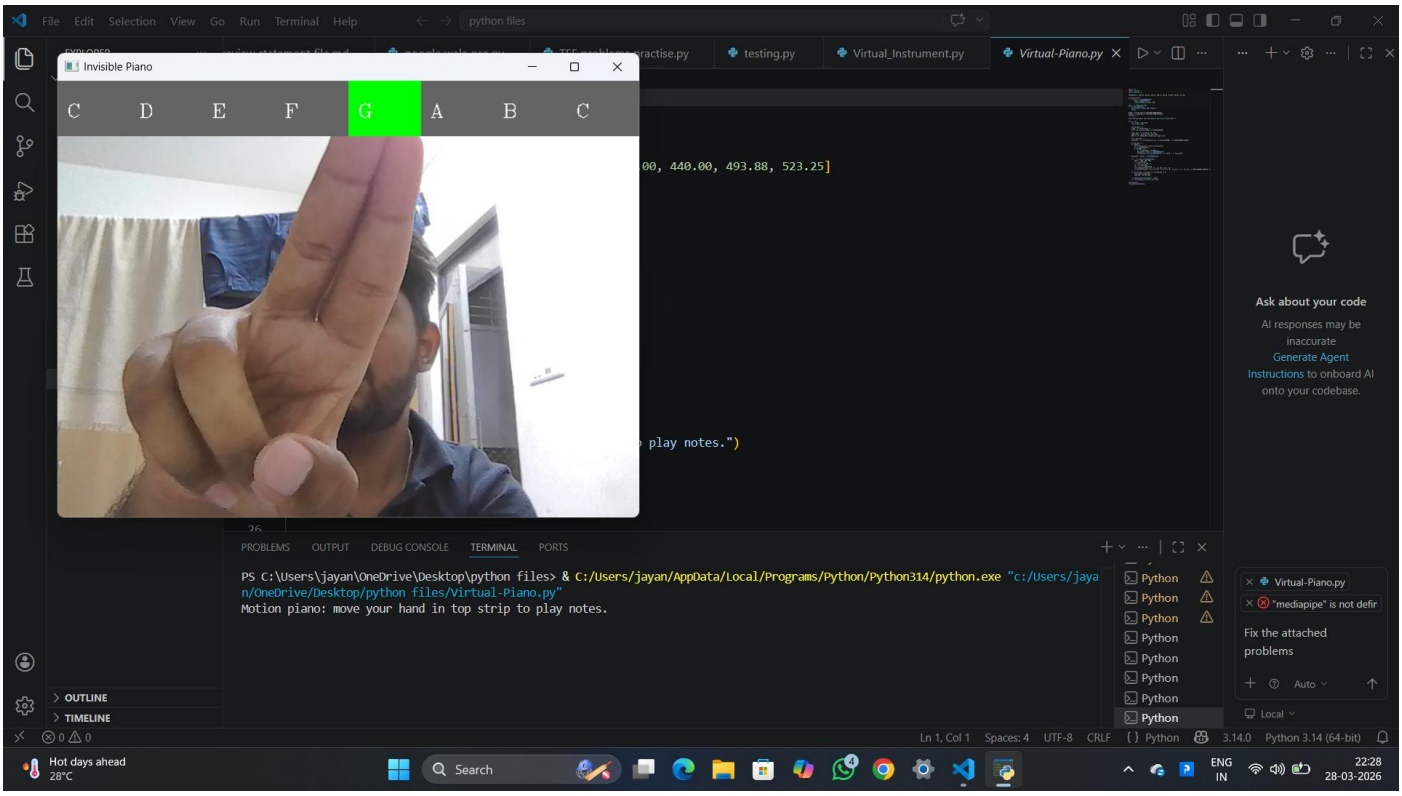
Motion piano: move your hand in top strip to play notes.
Ask about your code
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Fix the attached problems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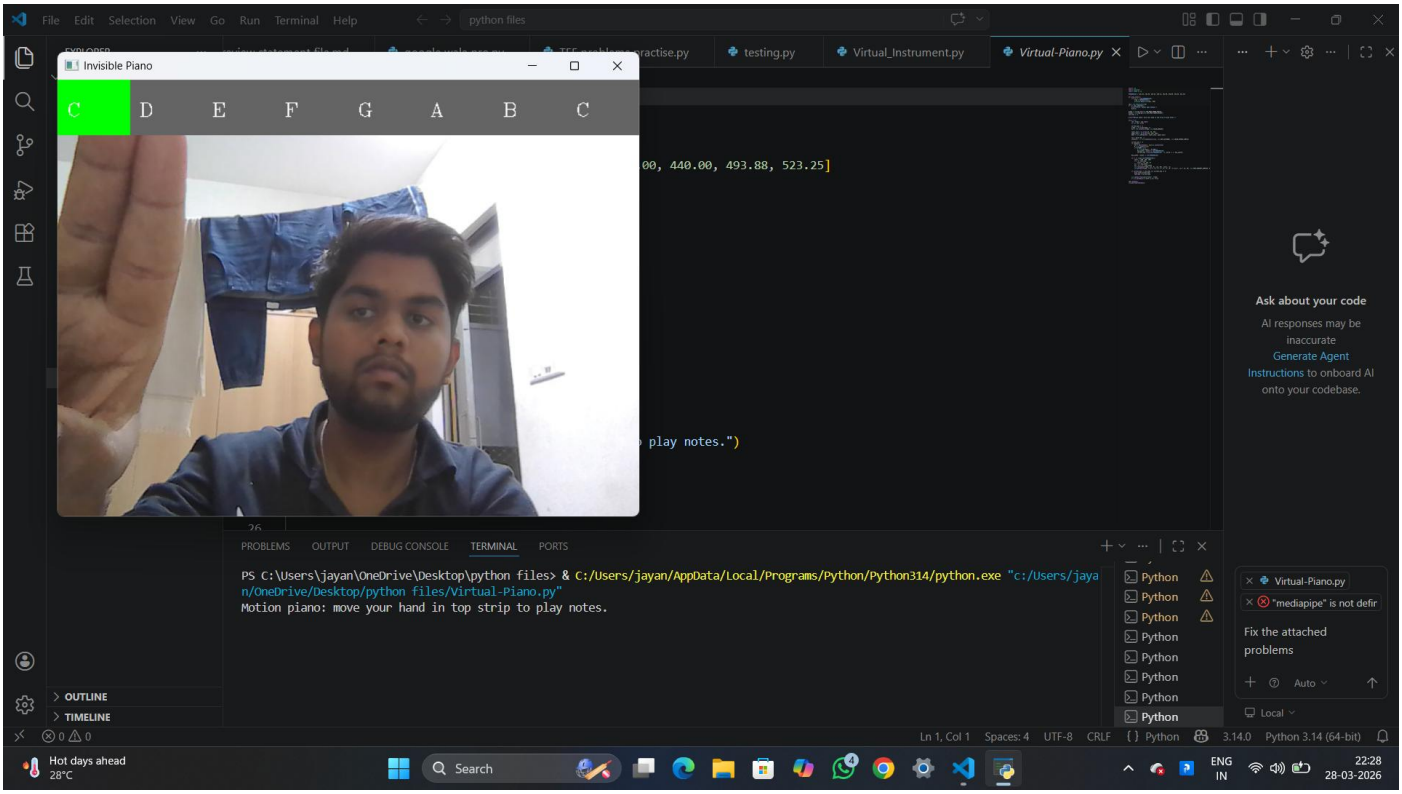
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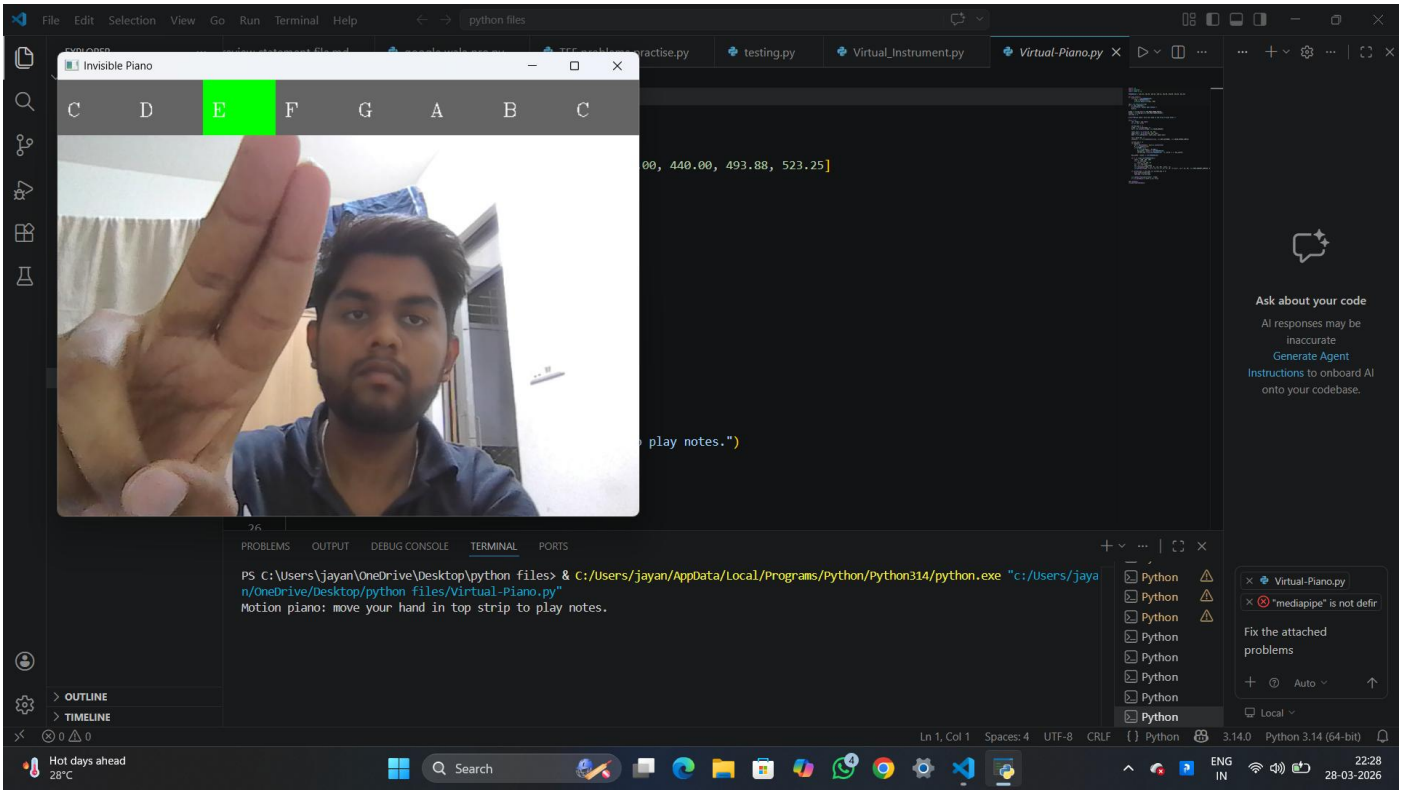
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6. Conclusion-

The project successfully demonstrates a functional gestural interface. By combining spatial partitioning with color-based object tracking, it creates an

intuitive and accessible virtual instrument.

THANK YOU.